



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

ADVANCED 5G/ 6G WIRELESS COMMUNICATIONS

DATA HANDLING AND HARDWARE PROGRAMMING

AT

FUTURE WIRELESS COMMUNICATIONS,

INDIAN INSTITUTE OF TECHNOLOGY – HYDERABAD

Internship Report

Submitted in Partial Fulfilment of the Requirement for

Awarding the Degree

Of

BACHELOR OF TECHNOLOGY

In

AERONAUTICAL ENGINEERING

By

Tirumala Sai Nithin

Roll Number: 20N31A2135

Employee ID: FWC22187

Under the esteemed guidance of,

Dr. G. V. V Sharma

Associate Professor

Electrical Engineering

Indian Institute of Technology Hyderabad

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CERTIFICATE OF COMPLETION

This is to certify that **Tirumala Sai Nithin**, a student of **Malla Reddy College of Engineering and Technology**, has successfully completed an internship at **Future Wireless Communications (FWC)**, Indian Institute of Technology (IIT) Hyderabad.

The internship was undertaken in the area of **advanced 5G/6G wireless communication**, with a focus on **Data Handling and Hardware Programming**. Throughout the internship period, he demonstrated commendable dedication, diligence, and technical proficiency in carrying out assigned tasks and projects.

This certificate is awarded in recognition of **Tirumala Sai Nithin's** valuable contributions, innovative ideas, and commitment to excellence during their tenure at FWC, IIT Hyderabad. We believe that the skills and knowledge acquired during this internship will serve as a solid foundation for their future endeavors in the field of wireless communications and beyond.

Dr. G. V. V. Sharma

Associate Professor

Department of Electrical Engineering

Future Wireless Communications (FWC)

Indian Institute of Technology (IIT) Hyderabad

ACKNOWLEDGEMENT

I would like to extend my sincere appreciation to all those who have played a crucial role in the successful completion of my internship report at the Future Wireless Communication (FWC), IIT Hyderabad.

Firstly, I am deeply grateful to **Dr. G. V. V. Sharma**, my internship head, for their unwavering guidance, support, and encouragement throughout this enriching experience. Their expertise and mentorship have significantly contributed to my professional development in **Data Handling and Hardware Programming**.

I would also like to express my gratitude to the entire team at FWC for providing me with an inspiring and collaborative environment to learn and thrive. Their cooperation and assistance have been indispensable in the accomplishment of my research objectives.

Furthermore, I extend my heartfelt thanks to the HR department of the company for their assistance in facilitating my internship placement and ensuring a smooth transition into the organization.

I am also indebted to the principal, faculty members, and staff of **Malla Reddy College of Engineering and Technology** for providing me with the opportunity to pursue this internship. Their support and encouragement have been instrumental in broadening my horizons and gaining practical insights into the field of [mention your field of study].

Last but not least, I am immensely grateful to my family and friends for their unwavering support and encouragement throughout this journey.

Thank you all for your invaluable contributions and belief in my abilities.

By,
Tirumala Sai Nithin,
FWC22187.

DECLARATION

I, **Tirumala Sai Nithin**, hereby declare that the internship report entitled "**Data Handling and Hardware Programming**" is an authentic representation of my work carried out during my internship at Future Wireless Communications (FWC), IIT Hyderabad, in the field of advanced 5G/6G wireless communication under the domain of Data Handling and Hardware Programming.

I affirm that the content presented in this report is the result of my independent research and analysis conducted under the supervision of **Dr. G. V. V. Sharma**. Wherever external sources have been utilized, proper citations and references have been provided to acknowledge the original authors and sources.

I further declare that this report has not been submitted, in part or in full, for the fulfillment of any other degree or qualification at any institution. Any similarities found with the work of others are purely coincidental and unintentional.

I take full responsibility for the accuracy, authenticity, and originality of the content presented in this report. Any opinions expressed herein are solely my own and do not necessarily reflect the views of Future Wireless Communications (FWC), IIT Hyderabad, or any other affiliated organization.

I understand the consequences of academic dishonesty and plagiarism and affirm that all ethical standards and guidelines have been adhered to throughout the preparation of this report.

Place: Hyderabad

By,
Tirumala Sai Nithin,
FWC22187.